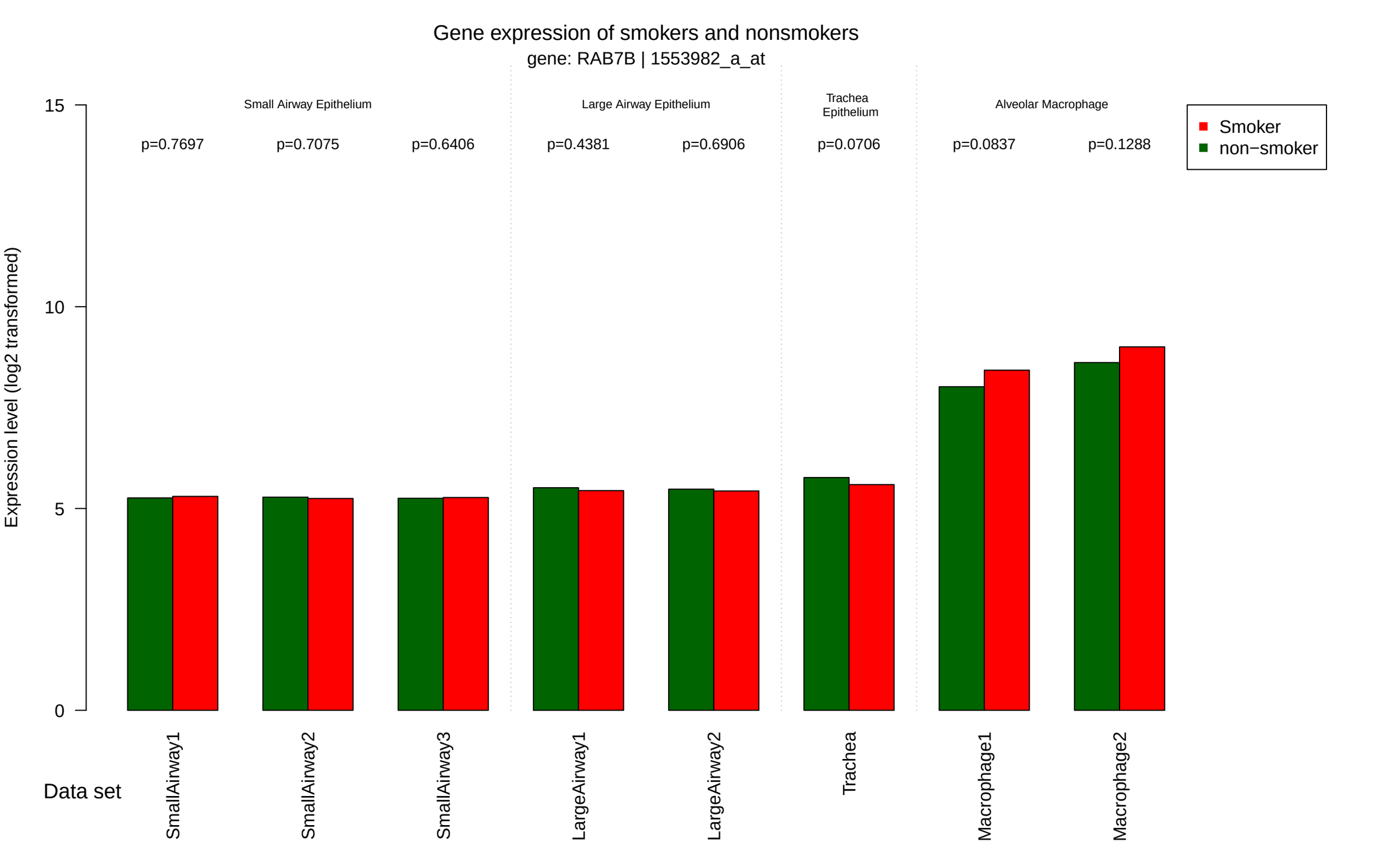
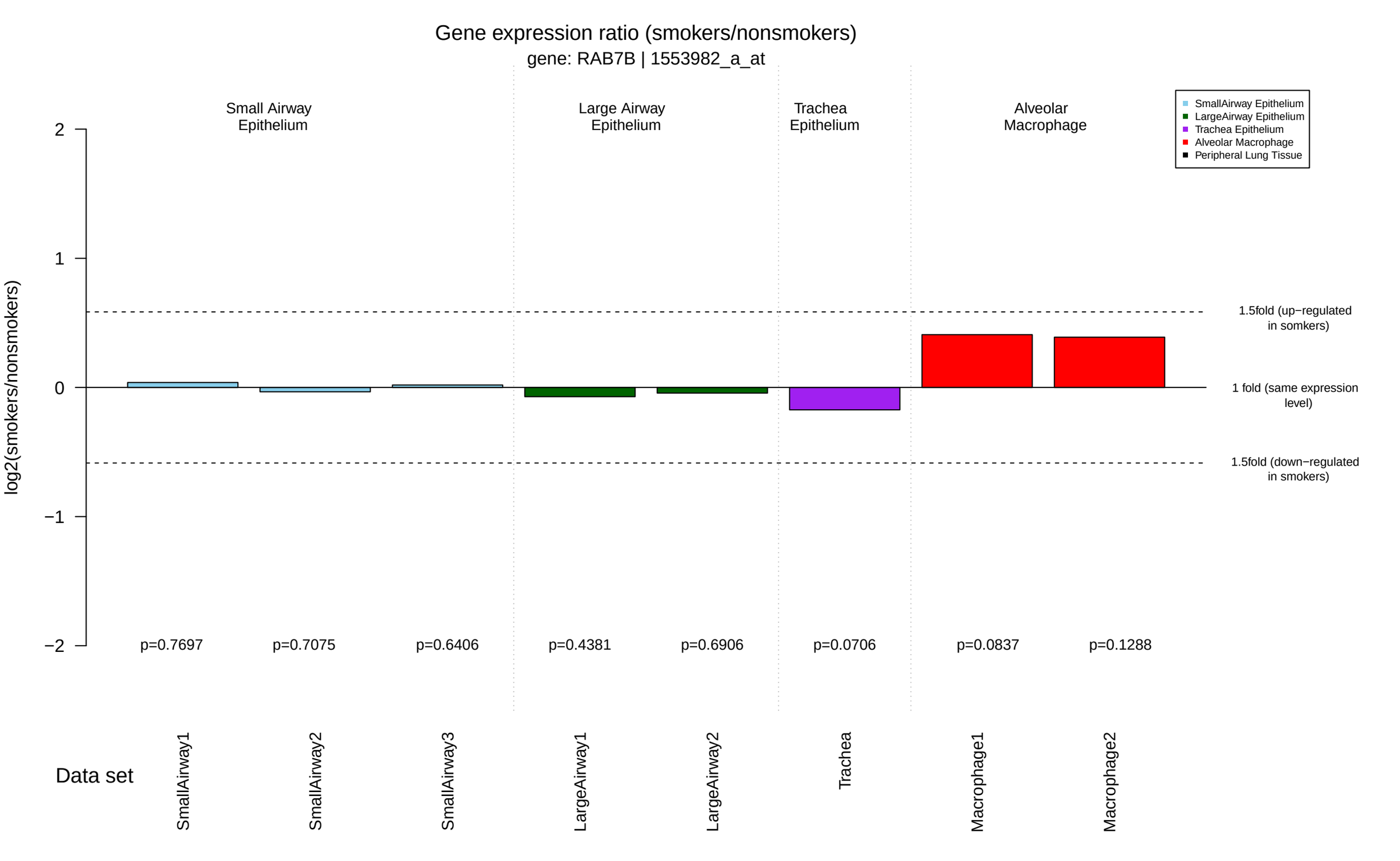
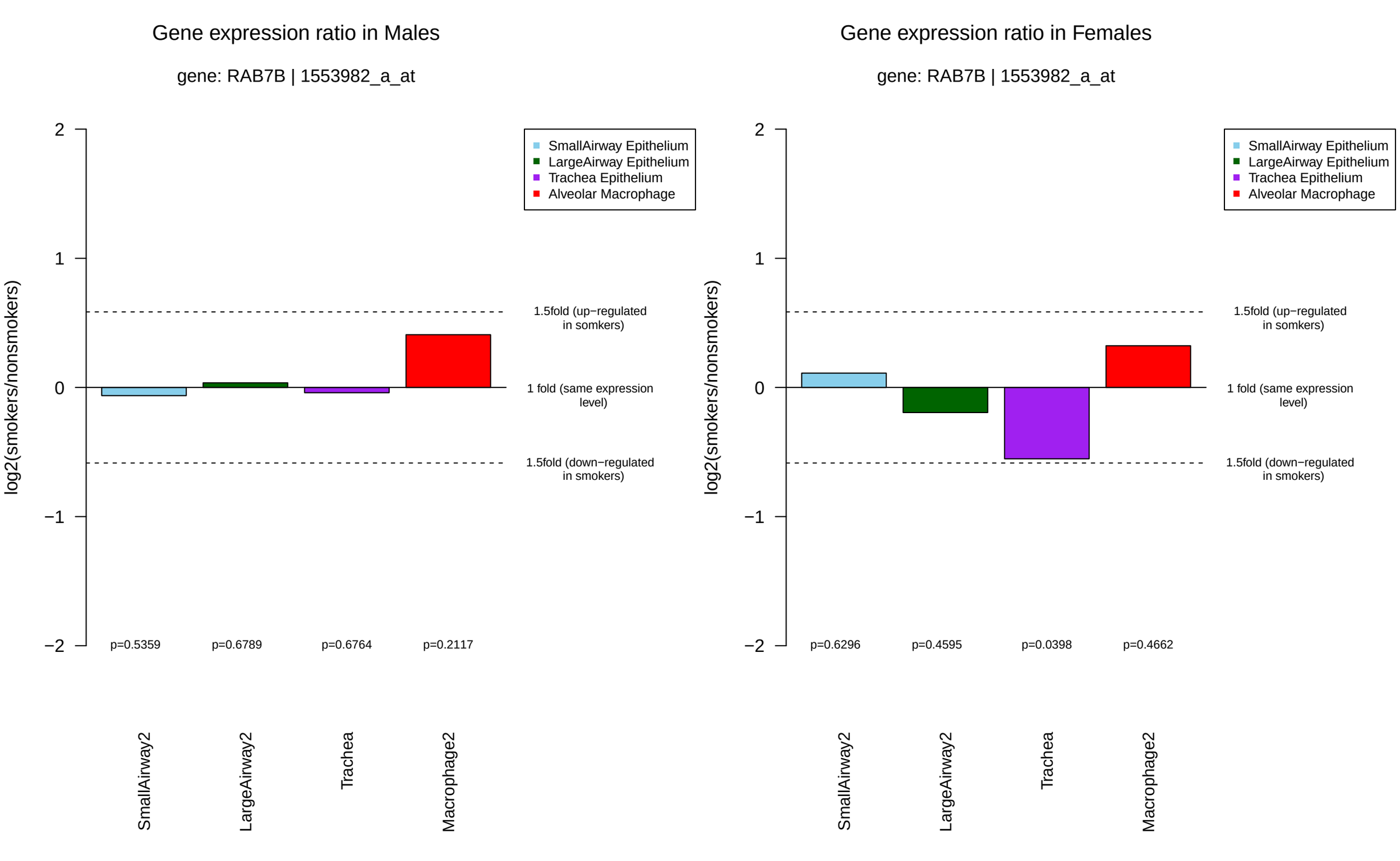
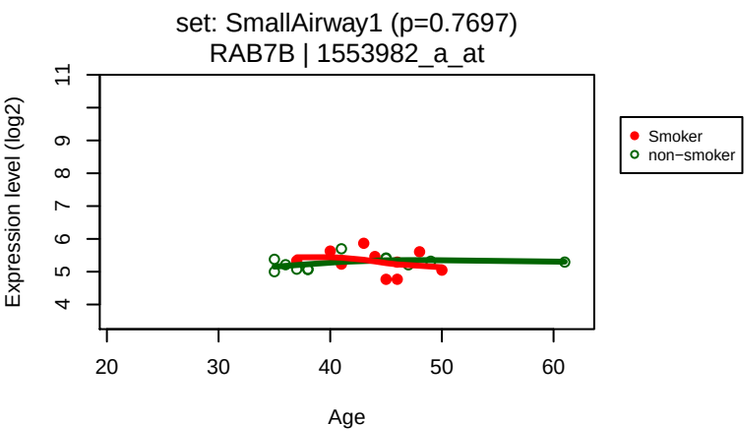
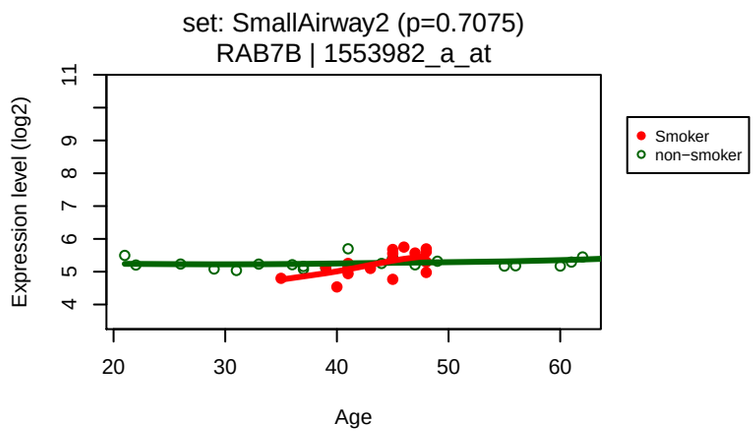
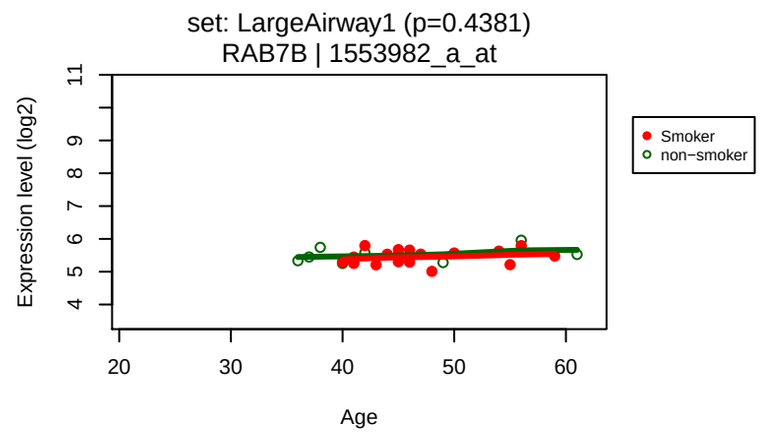

Gene expression of smokers and nonsmokers
gene: RAB7B | 1553982_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7697
p=0.7075
p=0.6406
p=0.4381
p=0.6906
p=0.0706
p=0.0837
p=0.1288
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

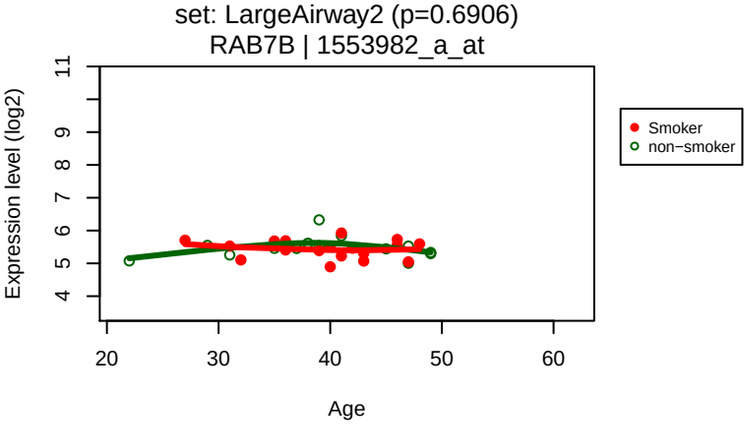

Gene expression ratio (smokers/nonsmokers)
gene: RAB7B | 1553982_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7697
p=0.7075
p=0.6406
p=0.4381
p=0.6906
p=0.0706
p=0.0837
p=0.1288
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

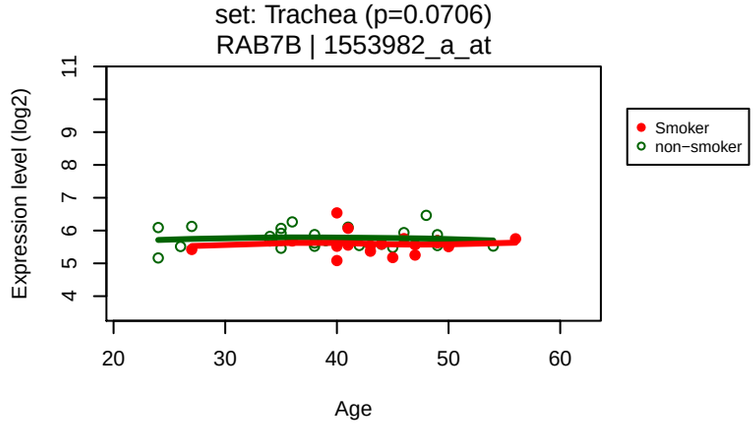

Gene expression ratio in Males
gene: RAB7B | 1553982_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5359
p=0.6789
p=0.6764
p=0.2117
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB7B | 1553982_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6296
p=0.4595
p=0.0398
p=0.4662
SmallAirway2
LargeAirway2
Trachea
Macrophage2

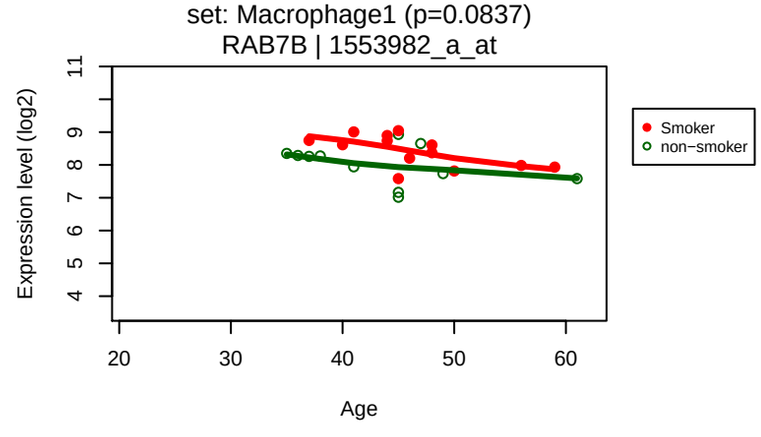

set: SmallAirway1 (p=0.7697)
RAB7B | 1553982_a_at
Expression level (log2)
Age
Smoker
non−smoker

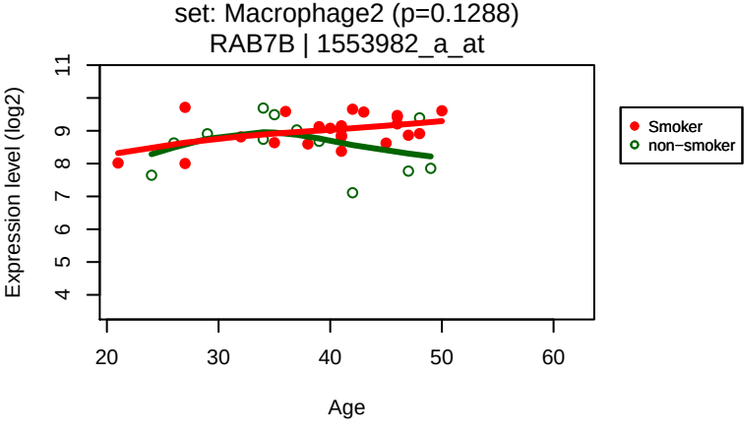

set: SmallAirway2 (p=0.7075)
RAB7B | 1553982_a_at
Expression level (log2)
Age
Smoker
non−smoker

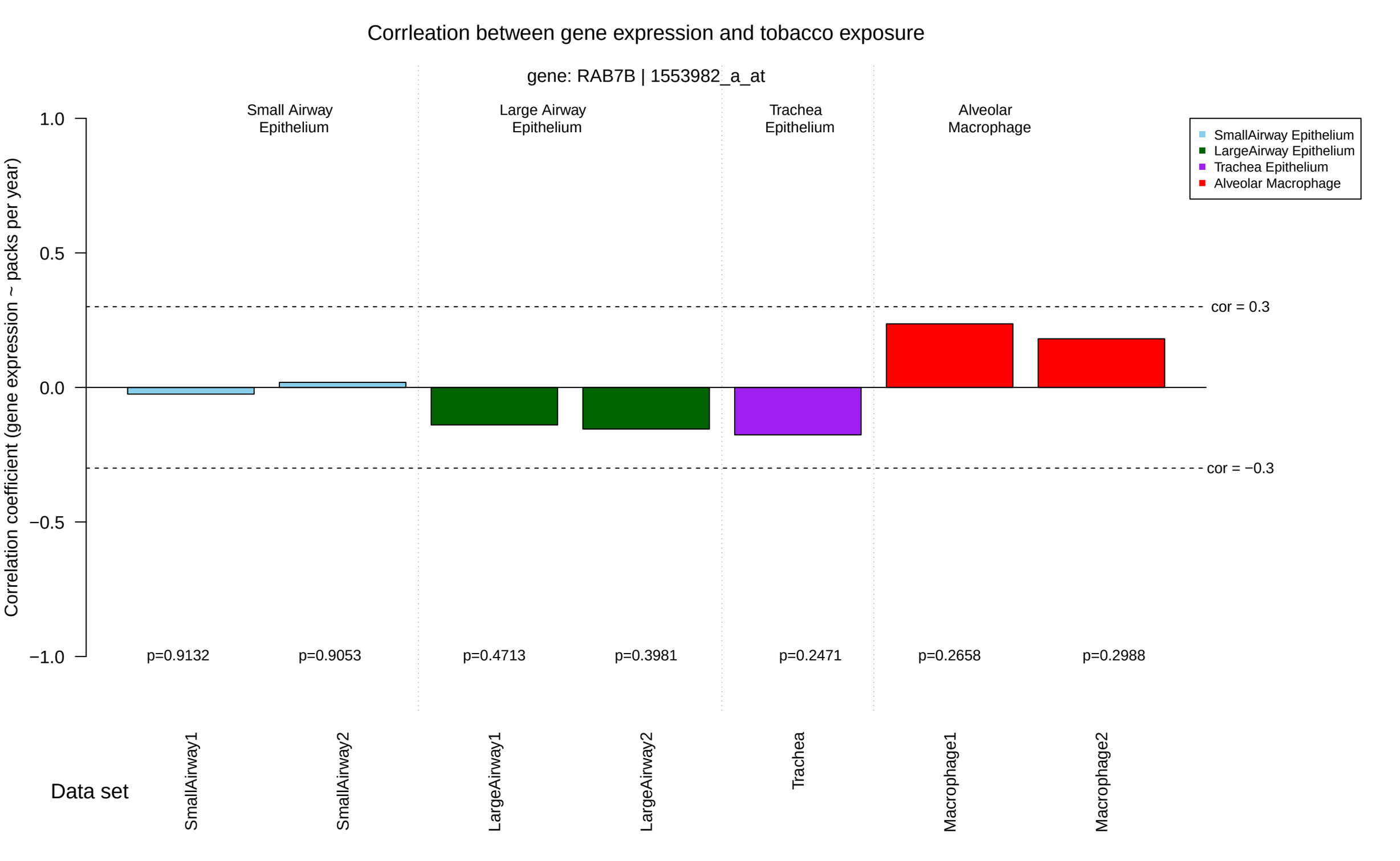

set: LargeAirway1 (p=0.4381)
RAB7B | 1553982_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6906)
RAB7B | 1553982_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0706)
RAB7B | 1553982_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0837)
RAB7B | 1553982_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1288)
RAB7B | 1553982_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB7B | 1553982_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9132
p=0.9053
p=0.4713
p=0.3981
p=0.2471
p=0.2658
p=0.2988
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2